SE2030 – Software Engineering
Lab Sheet 05 – Sequence Diagram

Project Title : Web-Based Vehicle Rental System - RideFlow

Group ID : 2026-Y2-S1-MLB-WEB3G2-38

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Project Type : Prorata – Individual Project

Use Case Scenario

Use Case : Manage Fleet & Vehicle Catalogue

Field	Description
Use Case ID	UC-02
Use Case Name	Manage Fleet & Vehicle Catalogue
Primary Actor	Rental Staff
Supporting Actor	Administrator
Goal	Maintain accurate vehicle information and availability status in RideFlow
Precondition	User is authenticated and has permission to manage vehicle records
Postcondition	Vehicle information is successfully created, updated, viewed, searched, or its status is changed
Trigger	Rental Staff selects the Fleet/Vehicle Management option

Main Success Scenario

1. Rental Staff logs into the RideFlow system.
2. The system authenticates the staff member and verifies their role.
3. Rental Staff opens **Fleet & Vehicle Catalogue Management**.
4. The system displays the existing vehicle records.
5. Rental Staff selects an operation such as **Add, View, Update, Search/Filter, or Manage Status**.
6. If adding a vehicle, the staff member enters the required vehicle information.
7. The system validates the entered information.
8. The system stores the vehicle record.
9. If updating a vehicle, the staff member selects an existing vehicle and modifies the required information.

10. The system validates the changes.
11. If the change requires administrator approval, the system sends the change for approval.
12. The administrator reviews and approves the change.
13. The system updates the vehicle record.
14. Rental Staff can search or filter vehicles using available criteria.
15. Rental Staff can update the vehicle's operational status.
16. The system saves the updated status.
17. The system displays the updated vehicle information.

Alternative Flows

A1 – Invalid Vehicle Information

1. Staff enters incomplete or invalid vehicle information.
2. The system validates the input.
3. The system identifies the invalid fields.
4. The system displays an error message.
5. Staff corrects the information.
6. The system validates the information again.

A2 – Vehicle Already Exists

1. Staff attempts to add a vehicle.
2. The system checks the vehicle information.
3. A matching vehicle record already exists.
4. The system prevents the duplicate record.
5. The system informs the staff member.

A3 – Administrator Rejects Important Change

1. Staff submits an important vehicle change.

2. The system sends the change for administrator approval.
3. Administrator reviews the change.
4. Administrator rejects the change.
5. The system keeps the previous information.
6. The system notifies the staff member.

A4 – Vehicle Under Maintenance

1. Staff attempts to make a vehicle available.
2. The system checks the vehicle's maintenance status.
3. The vehicle is currently under maintenance.
4. The system prevents the vehicle from being shown as available for rental.

Sequence Diagram

